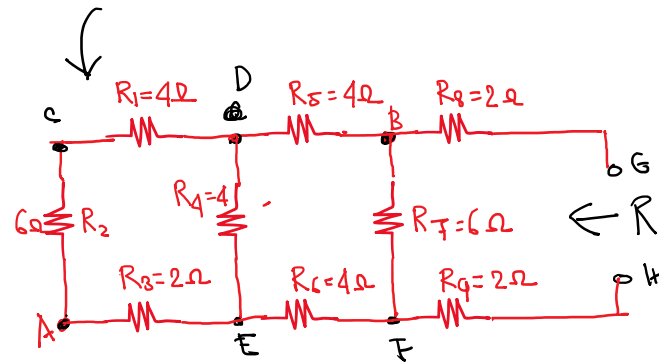
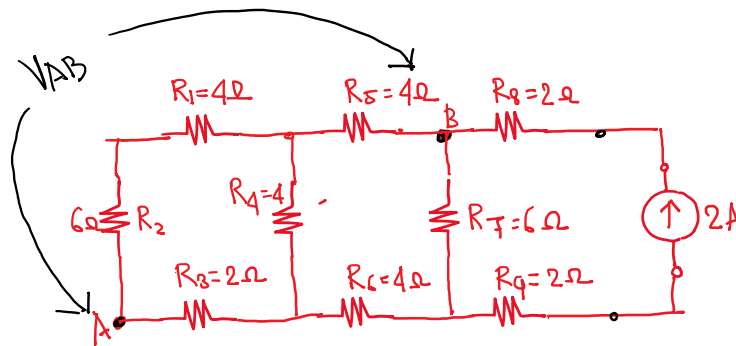
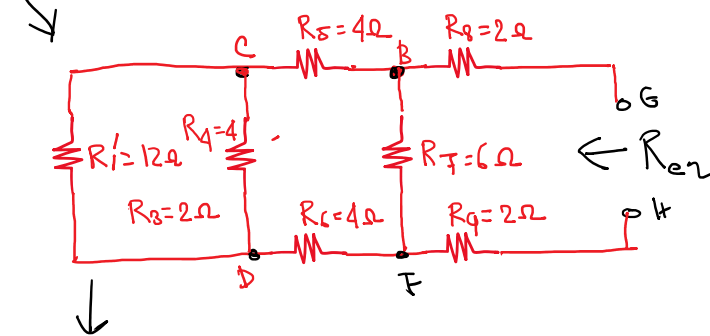




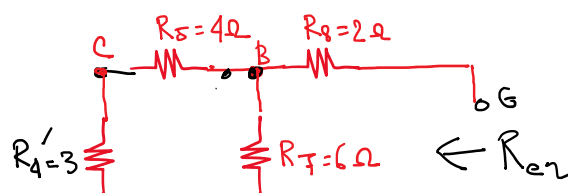
Remove the source by opening the terminals
← R_{eq} or nodes

Along DCAE $R_1 + R_2 + R_3 = R'_1 = 12\Omega$

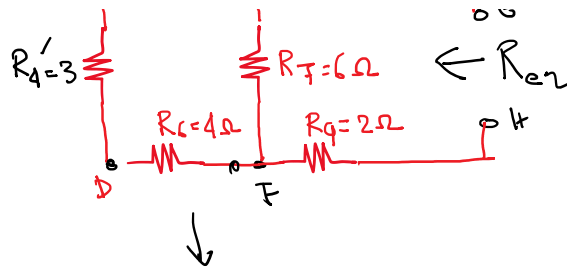


Between C & D R'_1 & R_4 are parallel

$$R'_4 = 12\Omega \parallel 4\Omega = 3\Omega$$

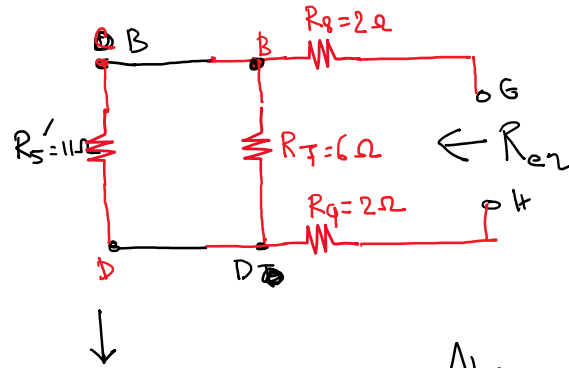


Along BCDF $R'_6 = R_5 + R'_4 + R_8$
 $= 4 + 3 + 2$
 $= 9\Omega$



$$= 4 + 3 + 4$$

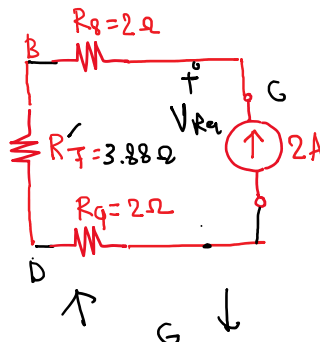
$$= 11\Omega$$



Between B & D $R_7' = R_5' \parallel R_7$

$$R_7' = \frac{66}{17} = 3.88\Omega$$

Redn



Along GBDH:

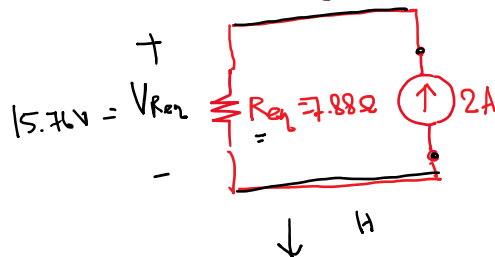
$$R_{eq} = R_8 + R_7' + R_9$$

$$= 2 + 3.88 + 2 =$$

$$= 7.88\Omega$$

$$I = \frac{V}{R_{eq}} \Rightarrow V = I R_{eq} = 2 \times 7.88$$

$$= 15.76V$$



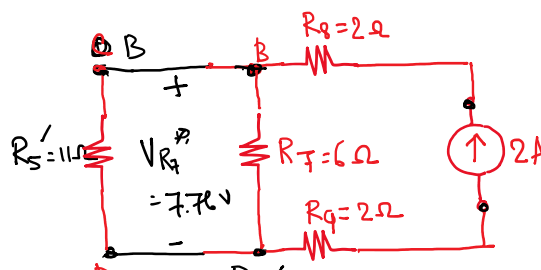
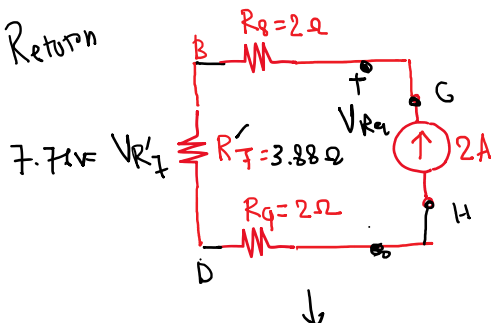
$$V_{R_5'} =$$

$$\frac{R_7'}{R_8 + R_7' + R_9} V_{Req}$$

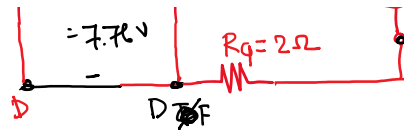
$$= \frac{3.88}{2 + 3.88 + 2} \times 15.76V$$

$$= 7.76V$$

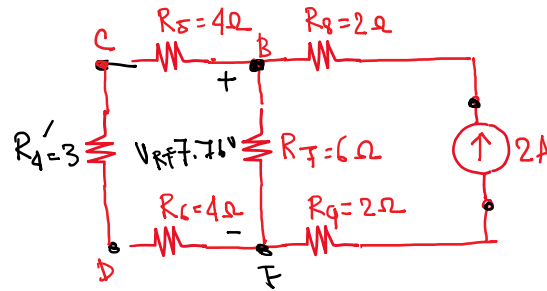
Return



$$V_{R_5'} = 7.76V = V_{R_7'}$$

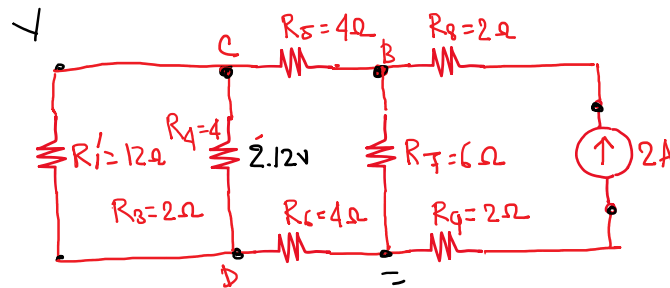


Between Node B and F/D

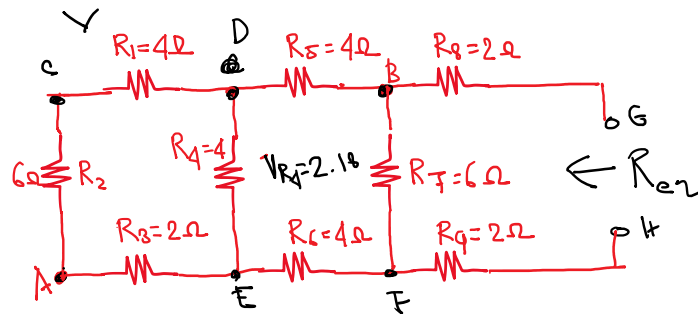


$$V_{R4}' = \frac{3}{4+3+4} \times 7.76 \text{ V}$$

$$= 2.12 \text{ V}$$

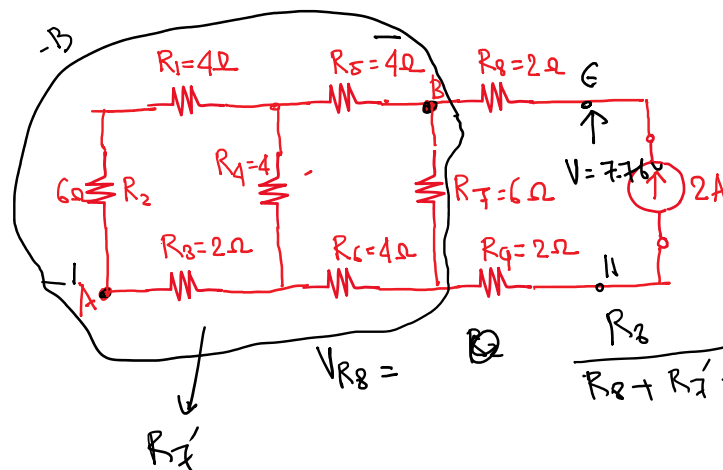


$$V_{R4} = V_{R'} = 2.12 \text{ V}$$

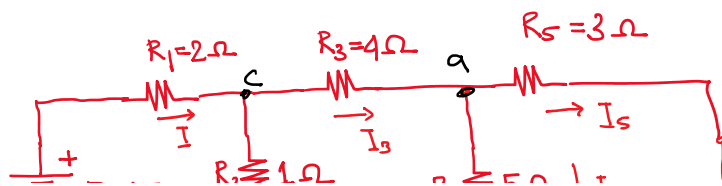


$$V_{R2} = \frac{R_2}{R_1 + R_2 + R_3} \times V_{R4}$$

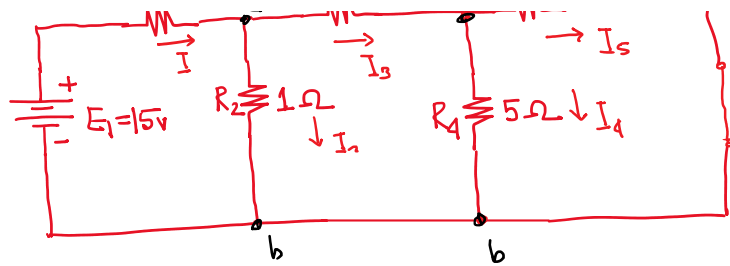
$$= \frac{6}{12} \times 2.12$$



$$V_{R8} = \frac{R_8}{R_8 + R_1' + R_7} \times 7.76$$

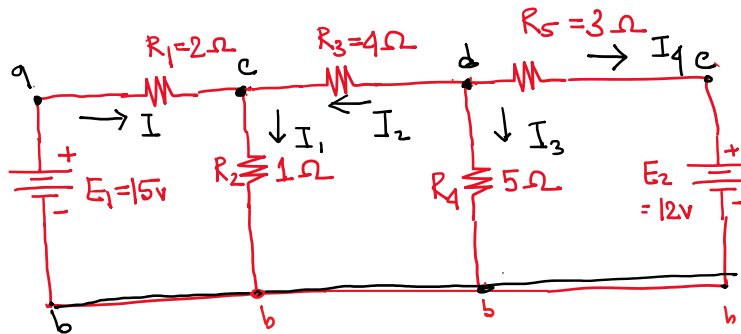


Reducible & Return method



$$R_{eq} = R_1 + R_2 \parallel [R_3 + (R_5 \parallel R_6)]$$

$$I = \frac{E}{R_{eq}}$$



I_5 Reducible and Return method valid

- ☐ Branch current method
- ☐ Node Analysis
- ☐ Mesh Analysis
- ☐ Superposition

☐ Branch current method

Steps: 1. Determine the number of branches in the ckt

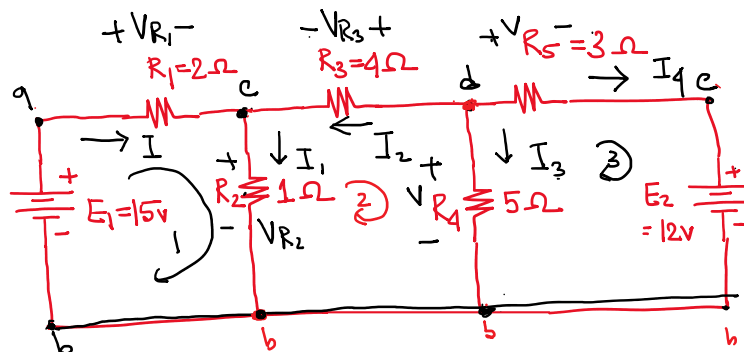
cab, cb, cd, db, bdc

Step 2: Assign current for each branch

Step 3: Apply KVL in all Independent loops.

$c \rightarrow a \rightarrow b \rightarrow c$: Independent loop: doesn't contain any other loop in it.

$d \rightarrow c \rightarrow a \rightarrow b \rightarrow d$: dependent loop. Why? it contains two loops in it.



KVL at Loop 1

$$+E_1 - VR_1 - VR_2 = 0$$

$$15 - IR_1 - IR_2 = 0$$

$$15 - 2I - 1I_1 = 0$$

$$\Rightarrow I_1 + 2I = 15 \quad \text{--- (I)}$$

$$\text{KVL @ Loop 2: } +VR_2 + VR_3 - VR_4 = 0$$

$$+ I_1 R_2 + I_2 R_3 - I_3 R_4 = 0$$

$$1I_1 + 4I_2 - 5I_3 = 0 \quad \text{--- (II)}$$

$$\text{KVL @ Loop 3: } +VR_4 - VR_5 - E_2 = 0$$

$$+ I_3 R_4 - I_4 R_5 - 12 = 0$$

$$\text{KVL @ Loop 3: } +V_{R4} - V_{R5} - E_2 = 0$$

$$+ I_3 R_4 - I_4 R_5 - 12 = 0$$

$$\Rightarrow 5 I_3 - 3 I_4 = 12 \quad \text{--- (III)}$$

Need two more equations. Apply KCL at minimum numbers of Nodes so that all the currents (I_1, I_2, I_3, I_4) are covered by the KCL.

KCL at c: I.C: $I + I_2$, o.C: I_1

$$I_1 = I + I_2 \Rightarrow I + I_2 - I_1 = 0 \quad \text{--- (IV)}$$

KCL @ d: I.C: 0, o.C: $I_2 + I_3 + I_4$

$$I_2 + I_3 + I_4 = 0 \quad \text{--- (V)}$$

KCL @ b: I.C: $I_1 + I_3 + I_4$, o.C: $I - V_1$

$$I_1 + I_3 + I_4 - I = 0$$

Redundant
No need

IV & V: Covers all the currents

$$I_1 + 2I = 15 \quad \text{--- (I)}$$

$$1I_1 + 4I_2 - 5I_3 = 0 \quad \text{--- (II)}$$

$$5I_3 - 3I_4 = 12 \quad \text{--- (III)}$$

$$I + I_2 - I_1 = 0$$

$$I_2 + I_3 + I_4 = 0$$

$$2I + I_1 + 0I_2 + 0I_3 + 0I_4 = 15$$

$$0I + 1I_1 + 4I_2 - 5I_3 + 0I_4 = 0$$

$$0I + 0I_1 + 0I_2 + 5I_3 - 3I_4 = 12$$

$$I - I_1 + I_2 + 0I_3 + 0I_4 = 0$$

$$0I + 0I_1 + I_2 + I_3 + I_4 = 0$$

$$\begin{bmatrix} 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 4 & -5 & 0 \end{bmatrix} \begin{bmatrix} I \\ I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix} = \begin{bmatrix} 15 \\ 0 \\ 12 \\ 0 \\ 0 \end{bmatrix}$$

Cramer's Rule: I_1, I_2, I_3